

QUESTIONS 18 / 18 completed

Natural

NATURAL

Write a program that prints the first n natural numbers using a for loop.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7         printf("%d",i);
8     return 0;
9 }
```

INPUT

6

QUESTIONS 18 / 18 completed

Sum

SUM

Write a program that calculates the sum of the first n natural numbers using a while loop.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i=1,sum=0;
5     scanf("%d",&n);
6     while(i<=n)
7     {
8         sum+=i;
9         i++;
10    }
11    printf("sum=%d",sum);
12    return 0;
13 }
14
```

INPUT

3

QUESTIONS 18 / 18 completed

Table

TABLE

Write a program that prints the multiplication table of a given number using a for loop.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,mul;
5     scanf("%d",&n);
6     for(i=1;i<=10;i++)
7     {
8         mul=n*i;
9         printf("%d*d=%d\n",n,i,mul);
10    }
11    return 0;
12 }
```

INPUT

5

QUESTIONS 18 / 18 completed

Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i=2,count=0;
5     scanf("%d",&n);
6     while(i<n)
7     {
8         if(n%i==0)
9             count++;
10        i++;
11    }
12    if(count==0)
13        printf("prime");
14    else
15        printf("not a prime");
16    return 0;
17 }
18
```

INPUT

11

QUESTIONS 18 / 18 completed

Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i=2,count=0;
5     scanf("%d",&n);
6     while(i<n)
7     {
8         if(n%i==0)
9             count++;
10        i++;
11    }
12    if(count==0)
13        printf("prime");
14    else
15        printf("not a prime");
16    return 0;
17 }
18
```

INPUT

11

QUESTIONS 18 / 18 completed

Count Vows

COUNT VOWS

Write a program that takes a string as input and counts the number of vowels in the string using a for loop.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     char str[100];
5     int i,count=0;
6     scanf("%[^\n]",str);
7     for(i=0;str[i]!=0;i++)
8     {
9         if(str[i]=='a' || str[i]=='e' || str[i]=='i' || str[i]=='o' || str[i]=='u' || str[i]=='A' || str[i]=='E' || str[i]=='I' || str[i]=='O' || str[i]=='U')
10        {
11            count++;
12        }
13    }
14    printf("%d",count);
15    return 0;
16 }
```

INPUT

apple

QUESTIONS 18 / 18 completed

Fibonacci

FIBONACCI

Write a program that prints the Fibonacci series up to a given number using a while loop.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,a=0,b=1,c,i;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         printf("%d",a);
9         c=a+b;
10        a=b;
11        b=c;
12    }
13    return 0;
14 }
```

INPUT

4

QUESTIONS 18 / 18 completed

Factorial

FACTORIAL

Write a program that takes a number as input and prints its factorial using a for loop.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,fact=1;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         fact*=i;
9     }
10    printf("%d",fact);
11    return 0;
12 }
```

INPUT

3

Pyramid

PYRAMID

Write a program that prints a pyramid pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,j;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=n-i;j++)
9             printf(" ");
10        for(j=1;j<=2*i-1;j++)
11            printf("* ");
12        printf("\n");
13    }
14    return 0;
15 }
```

INPUT

5

QUESTIONS 18 / 18 completed

Diamond

DIAMOND

Write a program that prints a diamond pattern of stars using nested loops.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,j;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=n-i;j++)
9             printf(" ");
10        for(j=1;j<=2*i-1;j++)
11            printf("* ");
12        printf("\n");
13    }
14    for(i=n-1;i>=1;i--)
15    {
16        for(j=1;j<=n-i;j++)
17            printf(" ");
18        for(j=1;j<=2*i-1;j++)
19            printf("* ");
20        printf("\n");
21    }
```

INPUT

5

QUESTIONS 18 / 18 completed

R- triangle ▾

R- TRIANGLE

Write a program that prints a right-angled triangle pattern of stars using nested loops.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,j;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=i;j++)
9             printf("* ");
10        printf("\n");
11    }
12    return 0;
13 }
```

INPUT

5

QUESTIONS 18 / 18 completed

Hollow Square ▾

HOLLOW SQUARE

Write a program that prints a hollow square pattern of stars using nested loops.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,j;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=n;j++)
9             if(i==1||i==n||j==1||j==n)
10                printf("* ");
11        else
12            printf(" ");
13        printf("\n");
14    }
15    return 0;
16 }
```

INPUT

5

QUESTIONS 18 / 18 completed

Hollow Diamond ▾

HOLLOW DIAMOND

Write a program that prints a hollow diamond pattern of stars using nested loops.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,j;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=n-i;j++)
9             printf(" ");
10        for(j=1;j<=2*i-1;j++)
11            if(j==1||j==2*i-1)
12                printf("* ");
13        else
14            printf(" ");
15        printf("\n");
16    }
17    for(i=n-1;i>=1;i--)
18    {
19        for(j=1;j<=n-i;j++)
20            printf(" ");
21        for(j=1;j<=2*i-1;j++)
```

INPUT

5

Pascal's triangle ▾

PASCAL'S TRIANGLE

Write a program that prints a Pascal's triangle pattern using nested loops.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int i,j,n,coeff=1;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)
6     {
7         for(j=0;j<=n-i;j++)
8             printf(" ");
9         for(j=0;j<=i;j++)
10        {
11            if(j==0||i==0)
12                coeff=1;
13            else
14                coeff=coeff*(i-j+1)/j;
15            printf("%4d",coeff);
16        }
17        printf("\n");
18    }
19    return 0;
20 }
```

INPUT

5

QUESTIONS 18 / 18 completed

zigzag

ZIGZAG

Write a program that prints a zigzag pattern of stars using nested loops.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,j,n,m;
5     scanf("%d %d",&m,&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=m;j++)
9         {
10             if((i+j)%4==0||(i==2 && j%4==0))
11                 printf("*");
12             else
13                 printf(" ");
14         }
15         printf("\n");
16     }
17     return 0;
18 }
```

INPUT

3 5

QUESTIONS 18 / 18 completed

Spiral pattern ▾

SPIRAL PATTERN

Write a program that prints a spiral pattern of numbers using nested loops.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,j,n;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=n;j++)
9         {
10             if(i==1||i==n||j==1||j==n||i==3&&j>=2&&j<=4)
11                 printf("* ");
12             else
13                 printf(" ");
14         }
15         printf("\n");
16     }
17     return 0;
18 }
```

INPUT

5

QUESTIONS 18 / 18 completed

Armstrong

ARMSTRONG

Write a program to find the given number is Armstrong or Not.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,temp,rem,sum=0;
5     scanf("%d",&temp);
6     n=temp;
7     while(temp!=0)
8     {
9         rem=temp%10;
10        sum=sum+(rem*rem*rem);
11        temp=temp/10;
12    }
13    if(sum==n)
14        printf("armstrong number");
15    else
16        printf("Not an armstrong Number");
17    return 0;
18 }
```

INPUT

153

QUESTIONS 18 / 18 completed

Perfect



PERFECT

Write a program to find
the given number is
Perfect number or Not.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,sum=0;
5     scanf("%d",&n);
6     for(i=1;i<n;i++)
7     {
8         if(n%i==0)
9         {
10             sum+=i;
11         }
12     }
13     if(sum==n)
14         printf("perfect number");
15     else
16         printf("not a perfect number");
17     return 0;
18 }
```

INPUT

28